

Additional Data Source Research — 2026-08

Requested by: Tiago · 2026-08-16 · **Scope:** research-only. No code, no credentials, no spend, no provider signup. This is a decision-aid menu, not an integration plan.

0. The one strategic finding: extend, don't add

The single most important result of this pass is that **most of the high-value additions are already reachable through the existing provider, Massive** (`api.massive.com`), not through new vendors. Massive's catalog (<https://massive.com/docs/llms.txt>) exposes, under the same `MASSIVE_API_KEY` already in use:

- **Indices** (REST OHLC aggregates + snapshots + a real-time value websocket) — so VIX, sector indices, and a SPY benchmark are a **ticker add**, not a provider add.
- **Stocks fundamentals — Short Interest** (`GET /stocks/v1/short-interest` , "included in all Stocks plans", updated every 2 weeks, history to 2017-12-29; fields `short_interest` , `days_to_cover` , `avg_daily_volume` , `settlement_date`).
- **Stocks fundamentals — Short Volume** (`/stocks/v1/short-volume` , daily off-exchange / ATS short-sale volume).
- **Economy — Treasury Yields** (plus inflation and labor market, all Federal Reserve data).
- **Stocks — News** (with sentiment), analyst ratings (Benzinga), earnings (Benzinga), ETF fund flows (ETF Global), corporate events (Wall Street Horizon), NBBO last-quote / quotes / tick trades.

Honest caveat: the documented **free** plan is 5 req/min, end-of-day option aggregates, and no option bid/ask/Greeks/IV/OI (see `docs/OPTIONS_QUOTE_SOURCE_EVALUATION.md`). Whether each endpoint above is on the free tier must be verified at <https://massive.com/dashboard> before wiring — but **no new signup, key, or provider** is required, which changes the signal-per-cost-per-effort math for several items below.

1. Covered already (reference only — do not re-propose)

- Indicators: `lib/pine_replay_v12_6_19.py` (RSI, EMA ladder + slopes, ATR, Bollinger, volume ratio, `campaignScore` , phase machine, 14 campaign events, `exhaustion_warning`).
- Sources wired: Massive free OHLC (intraday/daily) + options chain + option daily bars, TradingView webhooks, underlying/option heartbeats, SEC filings (`sec_filings.py`), FINRA daily short-sale volume (`finra_short.py`).
- System: paper execution (4 evidence roots, kill switch,

`AUTO_IF_VERY_HIGH_PAPER`), `dna_insight_library.py` closed vocabulary,

per-asset reliability mask, campaign lifecycle detector.

- Already-known gaps (reference, not new): real-time option bid/ask,

Greeks/delta-equivalent exposure, CTB/locates, per-position SL/TP brackets,

StochRSI divergence.

2. Evaluation tables by family

Columns: **what it adds** (which DNA decision + how it folds in — as a discrete flag / veto / carry cost / closed-vocabulary modifier, never a new score),

source · cost · credential, freshness (LIVE / DELAYED / STALE),

effort (small = client module + field; medium = new table/endpoint; large = new provider + backfill), **discipline fit, priority**.

2.1 Order flow / microstructure

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
Off-exchange / ATS ("dark-pool") volume + short volume	Off-exchange footprint read → closed-vocab modifier (off-exchange share bands) for the AMC campaign-lifecycle / squeeze reads	FINRA OTC (ATS & Non-ATS) Transparency — free, delayed, requires accepting a data agreement (https://www.finra.org/filing-reporting/otc-transparency , data at otctransparency.finra.org); or Massive /stocks/v1/short-volume (same key)	DELAYED (T+1)	Small–Medium	OK	Medium
Time & sales / tick tape	Microstructure burst / print flags → discrete flag	Massive/Polygon trades endpoint (real-time = paid tier)	LIVE (paid)	Medium–Large	OK	Reject (cost)
Footprint / volume profile	Structural level evidence (could inform Trade Box levels)	Derived from tick data (paid), or low-fidelity from existing bars	DELAYED	Large	OK	Reject (cost)
Order-book depth / balance (L2)	Depth-imbalance flag → entry timing	Broker feed or paid L2; no zero-credential source	LIVE (paid)	Large	Reject — requires broker/feed	Reject
Effective spread	Execution-quality realism	NBBO quotes (real-time = paid; options NBBO is a known gap)	DELAYED–LIVE	Small	OK	Defer (overlaps known gap)

2.2 Options flow

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
Index put/call ratio	Sentiment regime modifier (fear/greed) + veto on extremes → closed-vocab modifier	CBOE daily market statistics — free, public, no key (https://www.cboe.com/us/options/market_statistics/daily/)	DELAYED (daily EOD)	Small	OK	Medium

Per-ticker put/call	Asset-level sentiment	Derive from existing Massive option daily bars (covered data)	DELAYED	Small	OK	Low (re-derivation)
Unusual options activity	Entry-timing / conviction flag	Unusual Whales / FlowAlgo / Barchart — paid (~\$40–200+/mo)	DELAYED–LIVE	Medium	OK	Reject (cost)
GEX / gamma exposure	Dealer-hedging pin / acceleration read	SpotGamma (paid) or compute from OI + Greeks (free plan has neither)	DELAYED	Large	Reject — would synthesize Greeks	Reject (discipline)
IV rank / IV term structure	Vol-percentile regime → modifier	Needs option IV (free plan lacks IV) — known gap	DELAYED	Medium	OK	Defer (known gap)
Max pain	Pinning level → discrete flag	Needs OI by strike (free plan lacks OI) — known gap	DELAYED	Medium	OK	Defer (known gap)

2.3 Short / borrow (beyond the FINRA CDN already wired)

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
Consolidated short interest + days-to-cover	Squeeze / positioning read → closed-vocab modifier extending <code>SHORT_PRESSURE_RISK</code> in <code>dna_insight_library.py</code> , plus sizing context	Massive <code>/stocks/v1/short-interest</code> (in all Stocks plans, bi-weekly) — same key; alt: FINRA Query API (free registration, <code>developer.finra.org</code>)	DELAYED (bi-monthly)	Small–Medium	OK	High
Off-exchange / ATS short volume	Same slice as 2.1 row 1	Massive <code>/stocks/v1/short-volume</code>	DELAYED	Small	OK	Medium
CTB rate / shares available / locates	Borrow-cost carry + squeeze trigger	iborrowdesk.com (free web, no API, ToS-scrape) / ORTEX, S3 (paid); locates are broker-only	DELAYED	Medium	Reject — ToS scrape / broker locates	Reject
Utilization / borrow fee	—	ORTX / S3 (paid)	DELAYED	Medium	OK	Reject (cost)

2.4 Liquidity / execution

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
ADV bands	Position-sizing / carry-cost modifier + liquidity veto	Derive from existing Massive daily OHLC (zero new source)	DELAYED (daily)	Small	OK	High
Bid-ask spread (stocks)	Execution realism for shares	Massive stocks NBBO last-quote (free-tier TBD)	LIVE (if included)	Small	OK	Medium
Slippage model	Sizing realism	Needs spread + depth (see known gap)	—	Medium	OK	Defer (needs gap)

2.5 Sentiment / positioning

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
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News + sentiment	Catalyst flag → discrete flag	Massive news endpoint (likely paid tier)	LIVE	Medium	OK	Defer (cost/verify)
Social (Reddit / X / StockTwits)	Retail sentiment	Key-gated / paywalled / ToS-restricted APIs	LIVE	Medium	Reject — ToS/key	Reject
Analyst ratings / revisions	Consensus modifier	Massive Benzinga partner endpoints (likely paid) / Finnhub free tier (key)	DELAYED	Medium	OK	Defer (cost/key)
Insider transactions	—	Covered — <code>sec_filings.py</code> (INSIDER_FILING)	—	—	—	Covered
Buybacks	—	Covered via SEC filings + Massive corporate actions	—	—	—	Covered (Low enhancement)
Fund flows (ETF)	Rotation read	Massive ETF Global fund flows (likely paid)	DELAYED	Medium	OK	Defer (cost)

2.6 Macro / cross-asset

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
VIX / vol regime	Vol-regime modifier + high-vol veto → closed-vocab modifier + veto	Massive index OHLC (VIX) — same key, add ticker; alt: FRED VIXCLS	DELAYED—LIVE	Small	OK	High
Sector / rotation	Relative-strength modifier → closed-vocab modifier	Massive sector-ETF OHLC (XLF/XLE/XLK/...) — add tickers	DELAYED—LIVE	Small	OK	High
Correlation / β	Position-sizing modifier	Derive from existing Massive OHLC (asset vs SPY)	DELAYED	Small	OK	Medium
Rates (10y / 2y curve)	Macro carry / regime context	FRED <code>fredgraph.csv</code> (zero-key, verified: https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10); or Massive Treasury Yields	DELAYED (daily)	Small	OK	Medium
Seasonality	Descriptive context only	Derive from existing daily OHLC	STALE	Small	OK	Low (descriptive)
Earnings / dividend calendar	Event veto — avoid entries into earnings	Massive Corporate Events (Wall Street Horizon) / SEC	DELAYED	Small—Medium	OK	Medium

2.7 Alternative data

Candidate	What it adds	Source · cost · credential	Freshness	Effort	Discipline fit	Priority
Web traffic	—	Similarweb (paid)	STALE	Large	OK	Reject (cost/coverage)
App usage	—	data.ai (paid)	STALE	Large	OK	Reject
Satellite / foot traffic	—	RS Metrics (paid)	STALE	Large	OK	Reject

Consumer spending	—	Massive Fable Data (European panel — wrong geography for AMC)	STALE	—	Reject — irrelevant	Reject
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3. Shortlist — top "do next"

Best signal-per-cost-per-effort for **this** project (paper challenge, AMC-anchored, no real bid/ask, no broker):

1. VIX vol-regime + sector/SPY benchmark via existing Massive index OHLC.

Add index/sector tickers to `massive_ohlc.py` ; derive a closed-vocabulary vol-regime (LOW/NORMAL/ELEVATED/HIGH) + a relative-strength read. Folds in as a modifier and a high-vol veto. Zero new credential, small effort, touches entry timing, exit/protection, sizing, and the reliability mask.

2. **Consolidated short interest + days-to-cover (Massive `/stocks/v1/short-interest`).** Completes the short/borrow family right next to the brand-new `finra_short.py` , on the same API key. Folds in as an extension of the `SHORT_PRESSURE_RISK` modifier (squeeze/positioning). AMC is a heavily shorted name, so this is the highest-value **AMC-specific** add.

3. ADV bands derived from existing daily OHLC. Pure derivation, zero new source; a position-sizing / carry-cost modifier + liquidity veto. Cheapest possible improvement to sizing realism.

4. FRED treasury yields (10y/2y curve) via zero-key `fredgraph.csv` . Macro carry/regime context; DELAYED but zero-credential and verifiably fetchable.

5. CBOE index put/call ratio (daily, free, no key). A one-number sentiment regime modifier with a public, checkable source.

Runner-up: **earnings/dividend calendar** as an event veto (avoids entering a campaign add right into a binary event) — slightly more effort than the top five because it needs a calendar endpoint or partner data.

Single highest-priority add: the macro/cross-asset extension (item 1) — VIX vol-regime + sector/SPY benchmark — because it is the only candidate that is (a) reachable with zero new credential, (b) LIVE-adjacent rather than DELAYED, and (c) broad enough to touch entry, exit, sizing, **and** the reliability mask simultaneously.

4. Defer / reject list

Item	Verdict	Why
GEX / gamma exposure	Reject (most surprising)	The single most-cited "options flow" edge, but computing it honestly needs OI + Greeks; the free plan has neither, and synthesizing Greeks is explicitly out of bounds ("no invented numbers"). It fails the <i>*discipline*</i> test, not just cost.
CTB rate / shares available / locates (via iborrowdesk)	Reject	<i>*Looks*</i> free, but iborrowdesk is an undocumented scrape of a broker's website (ToS), and locates are broker-only. Fails discipline + freshness.
Order-book depth / balance (L2)	Reject	Requires a broker feed or an expensive market-data subscription; violates no-broker + no-spend.
Unusual options activity (UOA vendors)	Reject	Paywalled (\$40–200+/mo), and the free plan lacks the OI/IV needed to even approximate it independently.

Time & sales / footprint / tick tape	Reject	Real-time tick data is paid; low-fidelity derivations from existing bars are already covered by volume ratio / Bollinger.
Social sentiment (Reddit / X / StockTwits)	Reject	Key-gated, paywalled, or ToS-restricted; no honest zero-credential path.
IV rank / IV term structure / Max pain	Defer	Correct ideas, but each depends on IV or OI, both absent on the free plan (already-known gap). Revisit only after an options-quotes upgrade.
Effective spread / slippage model	Defer	Depends on real-time bid/ask (known gap). Stock-side NBBO via Massive is the cheap partial step.
Alternative data (web/app/satellite)	Reject	Paid and coverage-thin; classic cost/coverage traps with no paper-challenge payoff.
Seasonality	Low	Descriptive only; the project's evidence rules forbid promoting descriptive reads to decision inputs without a controlled study.

5. Boundary

No provider was contacted, no credential added, no money spent, no signup made, and nothing in `paper_execution/` or `massive_*.py` was modified. Every source claim above was checked against its documented page during this session (FINRA short-sale-volume catalog, FINRA short-interest reporting, FINRA OTC/ATS transparency, CBOE daily market statistics, FRED fredgraph CSV, and the Massive docs index). Free-tier eligibility of any Massive endpoint is a **pending verification**, not an assumption.